Multivariable Calculus: Integrals and Gradient Fields

1. Fundamental Theorem of Line Integrals

Consider a scalar potential function f and a smooth curve C parameterized by r(t). The line integral through gradient field satisfies:

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Where nabla f denotes the spatial gradient vector: $$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

2. Matrix Determinant and Eigenvalue Formulation

For an n x n transformation matrix A, characteristic polynomial solves:

$$\det(A - \lambda I) = 0$$